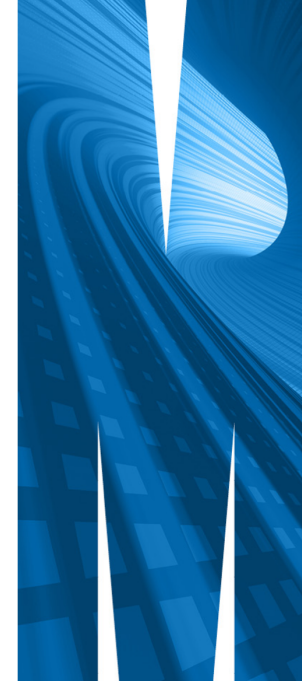
MONASH University

ETX1100/ETX5900 Business Statistics

Week 3: Sampling and confidence intervals

8 December 2026



Outline

1. From population to sample
2. Why statistics vary
3. Intervals for a mean
4. Intervals for a proportion

Today's question

What can one sample tell us about a population, and how uncertain is that estimate?

The lecture follows the ETF sequence: sampling design $\rightarrow$ sampling distribution $\rightarrow$ standard error $\rightarrow$ confidence interval. We apply the same logic to a mean and a proportion.

Outline

1 From population to sample

2 Why statistics vary

3 Intervals for a mean

4 Intervals for a proportion

The direction of inference

A **population** contains the units about which we want a conclusion. We observe a **sample**, compute a **statistic**, and use it to estimate a population **parameter**.

Population parameter	Sample statistic
Mean μ	Mean $\bar{x}$
Standard deviation σ	Standard deviation s
Proportion π	Proportion $\hat{p} = x/n$

A number in Excel is a sample statistic until the population and sampling method justify a broader claim.

Why sample selection matters

ETF's simple random sample gives each population member an equal chance of selection. The goal is to reduce **bias**, systematic favouring of some outcomes.

An in-app food-delivery poll misses nonusers. A train survey only at 8–9 am Friday may miss other travellers. A large sample from the wrong frame can still give an invalid population conclusion.

Watch me: diagnose a sample

A customer-list survey estimates a population defined by the list. Check:

1. Who is in the target population?
2. Who could actually be selected?
3. How were units selected?
4. Who did not respond or was not measured?
5. Could the recorded variable differ from the intended one?

This check belongs beside every interval, because a narrow numerical interval cannot repair selection or measurement bias.

Now you try: improve a survey

Time allowed: 6 minutes

A retailer emails only loyalty-card members to estimate all customers' satisfaction. Identify the target population, sampling frame and likely bias. Propose one practical improvement and state which limitation remains.

Outline

1 From population to sample

2 Why statistics vary

3 Intervals for a mean

4 Intervals for a proportion

The sampling distribution

ETF defines the **sampling distribution of** $\bar{x}$ as the distribution of all possible sample means from all possible samples of a fixed size n.

We observe only one mean, but another random sample would usually give another value. Across repeated samples, the mean of $\bar{x}$ is the population mean μ; the SD of $\bar{x}$ is the **standard error**:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}.$$

Shape and the Central Limit Theorem

If the population is Normal, the sample mean is Normal for any sample size. For a sufficiently large independent sample, the Central Limit Theorem makes the distribution of $\bar{x}$ approximately Normal even when the individual data are not Normal.

ETF uses $n \geq 30$ as a useful classroom guide, with a separate check for severe skewness or outliers. Larger n reduces SE because of the square root in the denominator.

Watch me: change sample size

Suppose σ = 24 units. Compare the standard error of a mean from n=36 and n=144.

The formulas are `24/SQRT(36)` and `24/SQRT(144)`. Quadrupling the sample size halves the standard error. It does **not** remove selection bias.

Now you try: separate spread from precision

Time allowed: 5 minutes

A sample SD is 18 units. Calculate $s/\sqrt{n}$ for `n=25` and `n=100`. Explain in a sentence why the sample SD and the standard error answer different questions.

Outline

1 From population to sample

2 Why statistics vary

3 Intervals for a mean

4 Intervals for a proportion

Point estimate plus margin of error

ETF calls the observed $\bar{x}$ a **point estimate**. Because it varies from sample to sample, report a range of plausible values for μ:

$$\text{confidence interval} = \text{estimate} \pm \text{critical value} \times SE.$$

The confidence level describes the long-run success of this sampling-and-interval procedure. For 95% confidence, 2.5% of the reference distribution lies in each tail.

The middle 95%

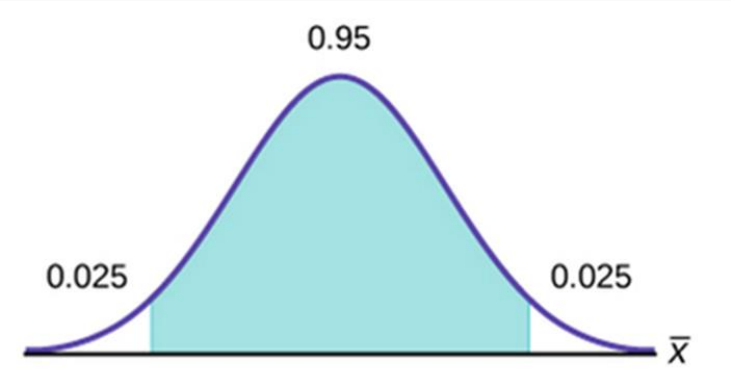


Figure 1: ETF confidence-area diagram: the central 95 percent and two 2.5 percent tails.

The diagram explains why Excel uses the **97.5th percentile** for

Known population SD: use z

When σ is genuinely known and the sampling conditions hold:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}, \qquad z^* = \texttt{NORM.S.INV}(0.975) \approx 1.960 \quad \text{for 95\% confidenc}$$

The units of the interval match the units of $\bar{x}$. The standard error becomes smaller as n grows.

Unknown population SD: use t

Usually σ is unknown, so ETF replaces it with the sample SD s and uses a t critical value with n-1 degrees of freedom:

$$\bar{x} \pm t^*_{n-1}\frac{s}{\sqrt{n}}, \qquad t^*_{n-1} = \texttt{T.INV}(0.975, n-1).$$

In Excel, `T.INV.2T(0.05,n-1)` gives the same positive 95% critical value. The t distribution's extra tail area reflects estimating the SD.

Watch me: an audit interval

ETF Workshop Week 5 asks about mean outstanding customer balances. A random sample has n=80 and $\bar{x}$ = \$86.05.

- If a historical **population** SD of \$22.38 is truly known, use $z^{*}\sigma/\sqrt{80}$.
- If only the **sample** SD \$24.02 is known, use $t^{*}_{79}s/\sqrt{80}$.

Build separate labelled cells for estimate, SD, n, critical value, SE, lower and upper bounds. The choice of critical value follows the source of the SD.

Now you try: two intervals, one question

Time allowed: 9 minutes

Use the audit figures in Week 3 estimation data. Construct both 95% intervals in Excel and explain which SD assumption each uses. Compare their widths without calling either interval a range for 95% of individual balances.

From raw rows to the mean interval

The `RealEstate` sheet in Week 3 proportions data holds 329 properties. For a price interval, do not start with an unlabeled supplied number:

- `n = COUNT(A2:A330)`
- $\bar{x}$ = *AVERAGE*(*A*2 : *A*330)
- `s = STDEV.S(A2:A330)`

Check `Price` units before writing the conclusion. If comparing school-zone groups, use group-specific data and denominators, and do not treat overlap of separate intervals as a formal test of a difference.

Watch me: change confidence and sample size

With the same point estimate and SD, change 95% to 90% and 99%. A higher confidence level gives a larger critical value and wider interval.

Then change n while holding the SD fixed. More observations reduce the standard error. These are **precision** effects; neither change fixes a biased sample.

Now you try: audit the conclusion

Time allowed: 6 minutes

Write a 95% mean-interval conclusion for a named population. Then correct this claim: “95% of the individual balances lie inside our confidence interval.” Explain which distribution the interval describes.

Outline

1 From population to sample

2 Why statistics vary

3 Intervals for a mean

4 Intervals for a proportion

A proportion estimates how common an event is

ETF uses π for a population proportion and $\hat{p} = x/n$ for the sample proportion, where x is the count with a named characteristic.

For a shopper outcome, define both the **event** and the **denominator**. $\hat{p}$ lies between 0 and 1. A sample proportion also varies across repeated samples.

The proportion standard error

For a large enough sample, the distribution of $\hat{p}$ is approximately Normal. For an interval, use the observed proportion in the estimated SE:

$$\widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

ETF's classroom check requires both $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$. These are counts of observed successes and failures. A rare event near the threshold deserves a cautious interpretation.

The 95% proportion interval

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \qquad z^* = \texttt{NORM.S.INV}(0.975).$$

The result is a proportion. Multiply endpoints by 100 to report percentages, and use the same denominator in the estimate and standard error. This familiar interval is an approximation, especially for very rare outcomes.

Watch me: Marketing from raw records

In `Marketing`, each row is a respondent and `Have Tried` is column E. ETF's seminar asks for the proportion who have tried the product.

Count `Yes` with `COUNTIF(E2:E857,"Yes")`, count respondents with `COUNTA(A2:A857)`, calculate $\hat{p} = x/n$, check `x` and `n-x`, then build the interval cell by cell. A PivotTable can check the count; the formulas make the interval auditable.

Now you try: change the population

Time allowed: 9 minutes

Estimate the proportion of **female respondents** who have tried the product. Filter or use `COUNTIFS` for the numerator and `COUNTIF` for the female denominator. Construct a 95% interval and write a sentence that names the female customer population, not all customers.

Interpreting 95% confidence

A defensible conclusion says: “Using this sampling procedure, we are 95% confident that the population proportion [or mean] is between the two reported bounds.”

It does **not** assign 95% probability to a fixed parameter after the sample is observed. It does **not** make the sample representative. State sampling, independence, approximation and measurement limitations alongside the number.

Closing workflow

Population and event $\rightarrow$ sampling design $\rightarrow$ count or mean from raw rows $\rightarrow$ standard error and conditions $\rightarrow$ critical value $\rightarrow$ interval $\rightarrow$ interpretation and limitation.

Next week, a hypothesis test asks whether the data provide enough evidence for a particular claim about that parameter.